SOLUTION CHALLENGE 2026 - BUILD WITH AI

GridSense AI

Bengaluru-Localized Smart Meter Intelligence & Revenue Recovery System for BESCOM

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- Theme 8: AI for Smart Meter Intelligence & Loss Detection by BESCOM

Problem Statement

Bengaluru utilities need fast, explainable loss detection from smart meter data.

Power theft and tampering create revenue leakage across dense Bengaluru feeder zones.

Evening peak demand in residential and industrial areas is hard to anticipate from raw readings alone.

Manual anomaly review is slow and can create false positives.

BESCOM officers need inspection priorities, not another raw-data dashboard.

12,480
smart meters simulated

18
high-risk meters

INR 2.4L
estimated loss detected

Brief About the Solution

GridSense AI converts Bengaluru smart meter patterns into demand, theft, and revenue decisions.

Forecast Demand

Predict hourly load and evening peak-risk windows for BESCOM zones.

Detect Theft

Flag abnormal drops, spikes, peer deviations, and tampering signals.

Recover Revenue

Rank inspection cases by confidence, loss impact, urgency, and tariff savings.

Opportunities and USP

Different from a normal dashboard: it explains and prioritizes action.

Typical dashboard

Shows historical readings; Requires manual anomaly search; Limited explainability; No revenue-priority workflow

GridSense AI

Predicts demand and grid stress; Detects theft-risk patterns automatically; Explains why a case was flagged; Creates BESCOM officer inspection priorities

Features Offered

Submission-ready capabilities demonstrated in the prototype.

Demand Forecasting

Actual vs predicted load with peak warning zone.

Theft Alerts

Critical, high, and medium risk meter anomalies.

Karnataka Data Layer

CSV-backed feeder profile with irrigation, solar, and evening domestic load patterns.

Bengaluru Zone Map

Peenya, Whitefield, KR Puram, Yelahanka, and Electronic City style zones.

Tariff + Carbon Engine

Monthly savings and CO_2 impact from BESCOM-style optimization scenarios.

Inspection Queue

Approve, reject, or escalate Bengaluru-area cases.

Explainability

Reason, formula, audit timeline, and officer summary prompt contract.

Process Flow Diagram

Smart meter data to BESCOM officer action in one auditable workflow.

Step 1

Smart meter data received

Step 2

AI predicts Bengaluru demand

Step 3

Anomaly detected

Step 4

Explanation generated

Step 5

Officer approves inspection

Step 6

Revenue recovery tracked

Architecture Diagram

Frontend-first, Vercel-safe decision-support layer on top of existing BESCOM systems.

Data Layer

Synthetic meter readings, Karnataka CSV load sample, zone load, peer groups

AI Layer

Precomputed forecasting, anomaly scoring, Gemini-ready explanation summaries

App Layer

React dashboard, tariff engine, scenario simulator, officer workflow

Cloud Layer

Vercel static prototype now; Firebase/GCP-ready storage for pilot

Technologies Used

Modern frontend prototype with Google AI-ready architecture.

- **React**
- **Vite**
- **Tailwind CSS**
- **Recharts**
- **Framer Motion**
- **Lucide Icons**
- **Vitest + GitHub Actions**
- **Gemini API / Google AI Studio ready**

Snapshots of the MVP

Core screens implemented in the live prototype.

Command Dashboard

Bengaluru impact metrics, simulate theft, live risk state

Demand + Karnataka Data

Actual vs predicted load plus localized feeder profile

AI Alerts + Summary

Meter risk, confidence, estimated loss, localized reason

Scenario Lab

Tariff savings, carbon impact, and officer actions

Implementation Cost and Future Development

Feasible as a staged pilot and scalable across zones.

Prototype

INR 0 cloud spend - static Vercel frontend with precomputed synthetic data

Pilot

Masked BESCOM connector, Gemini API, Firebase/GCP storage, officer login

Scale

Bengaluru circle rollouts, GIS map integration, model feedback loops

Future

Gemini assistant for officer case summaries

Future

Firebase ingestion pipeline for masked meter events

Future

Google Maps / BESCOM zone visualization

Future

Live BESCOM tariff and subsidy configuration management

Future

Model feedback loop from inspection outcomes

Submission Links and Demo Details

Ready for prototype submission. Replace placeholders after deployment/video upload.

GitHub Repository

https://github.com/Nikhilsai-9/gridsense-ai

Live Prototype

ADD_VERCEL_LINK_HERE

Demo Video

ADD_DEMO_VIDEO_LINK_HERE

Portfolio

https://nikhilsai-9.github.io/nikhilsai-portfolio/

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Prototype uses synthetic Bengaluru-style data only. Not an official BESCOM product. Built for hackathon demonstration and decision-support research.